Systems, Drawn · Video #5

Google Cloud Storage incident:
write errors in us-central1, 8–9 Oct 2026

Diagram pack · CC BY 4.0 · SRE // microservices

1. Two windows, never one bar: 12h30m and 5h41m
2. Normal behavior: capacity-aware routing (simplified)
3. 8 Oct: the traffic sorting optimization bypasses it
4. 8 Oct, step 1: emergency capacity, blocked
5. Same write, two routing paths
6. 8 Oct morning: start, detection, first status-page update
7. 8 Oct afternoon: last live update, mitigated, the four steps
8. 9 Oct: the recurrence, the reports, two ways to count the hours
9. The incomplete rollback: three names, three boxes

Source: Google Cloud Service Health, incident mSFVt6PiP557bjRyYsut (live updates 8 Oct, preliminary report 9 Oct, incident report 14 Oct 2026) and the Cloud Storage product history page. Times in PT (UTC−7 on these dates) as Google reports them; ART = PT + 4 h.

1. Two windows, never one bar: 12h30m and 5h41m



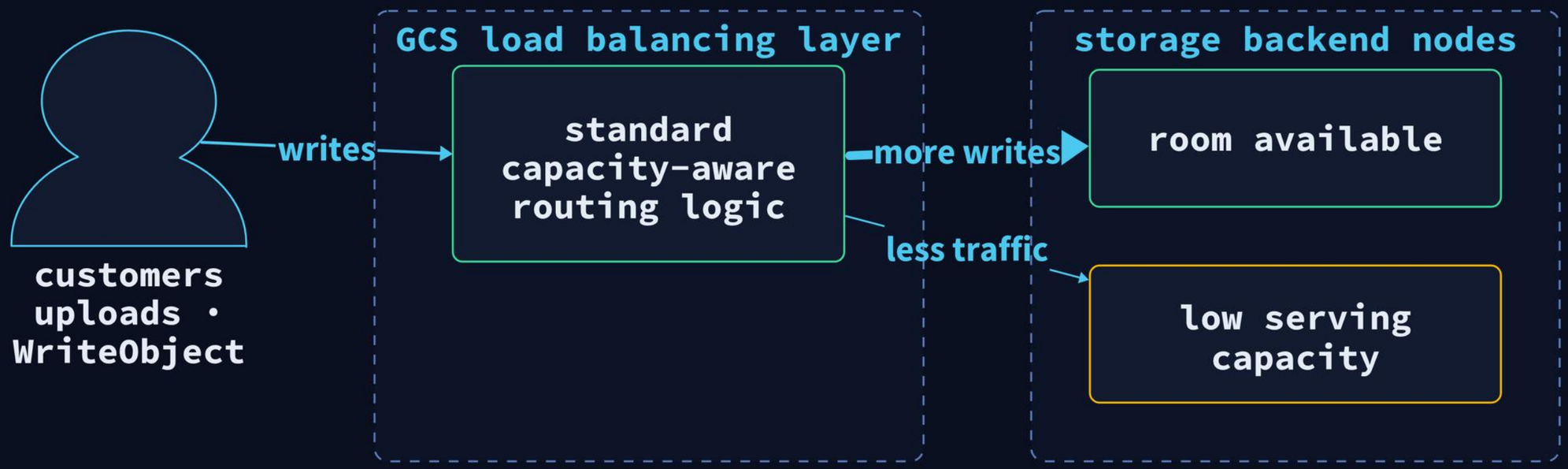
per Google's incident report

Systems, Drawn · CC BY 4.0 · src/two_windows_b01.d2

2. Normal behavior: capacity-aware routing (simplified)

Google Cloud Storage · us-central1 · normal behavior

simplified, based on Google's report

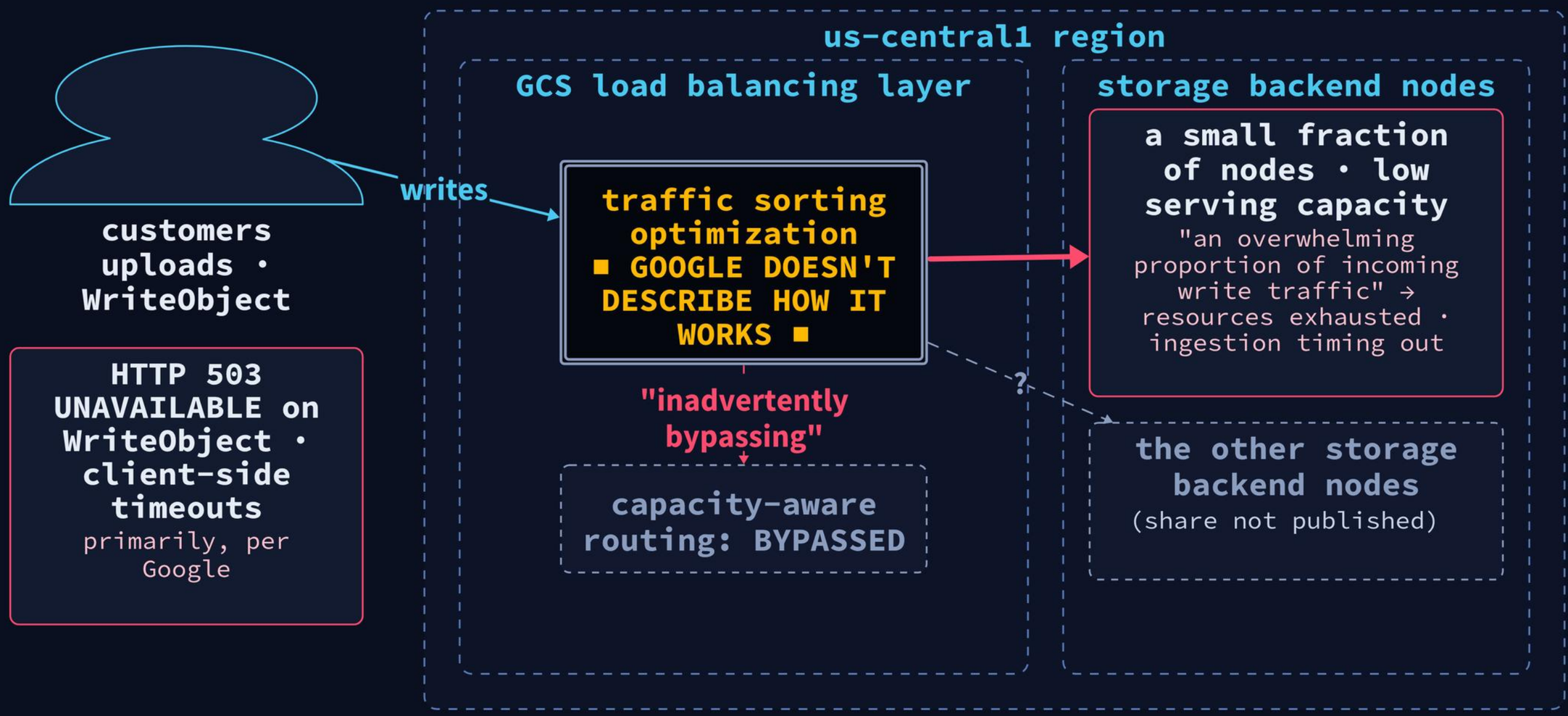


Google doesn't publish this topology or how the routing weighs capacity · node count and arrow weights are illustrative
Google, on the full revert: "restoring the previous capacity-aware routing behavior and allowing traffic to distribute evenly across the region" · incident report, 14 Oct 2026 · status.cloud.google.com/incidents/mSFVt6PiP557bjRyYsut

Systems, Drawn · CC BY 4.0 · src/topology_capacity_aware.d2

3. 8 Oct: the traffic sorting optimization bypasses it

Google Cloud Storage · us-central1 · 8 Oct 2026, 04:00–16:30 PT



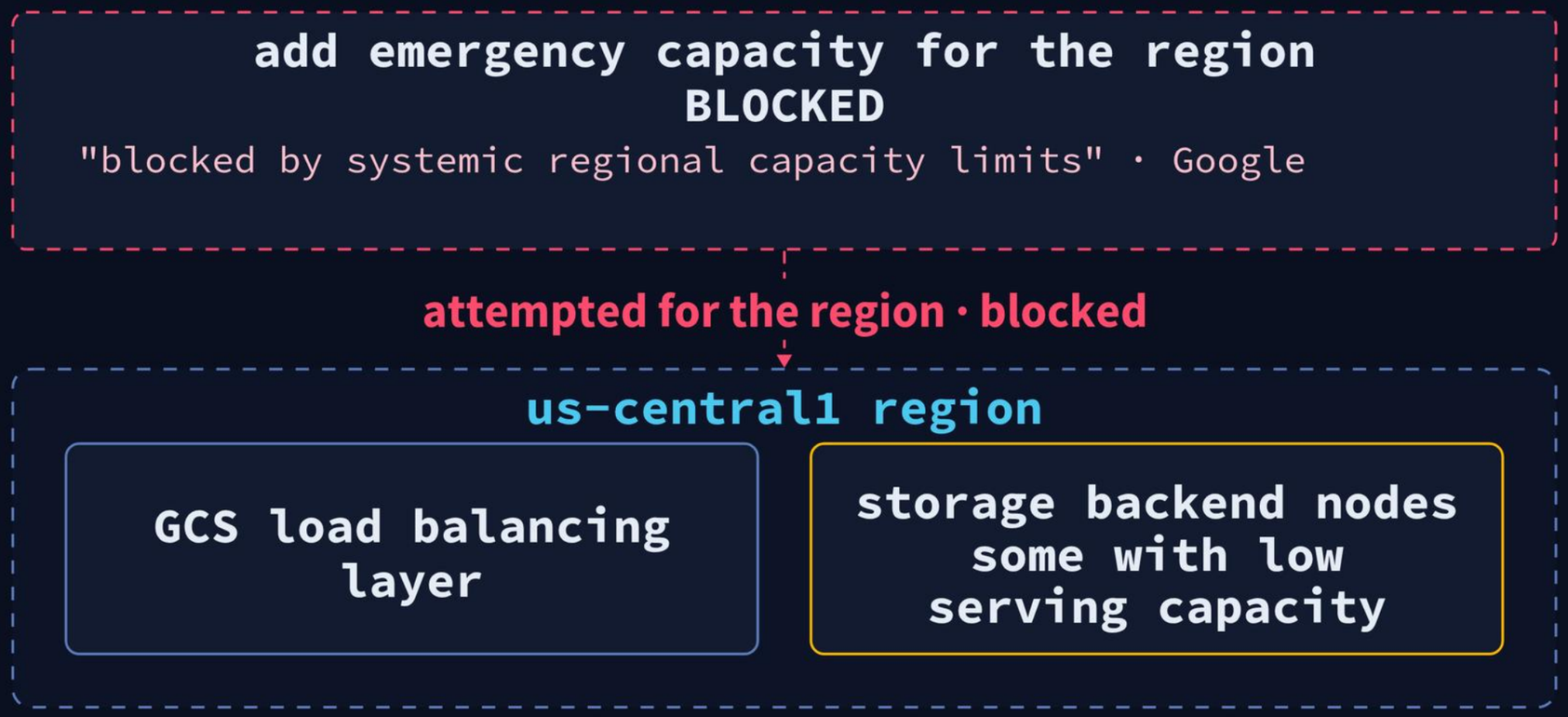
simplified, per Google's incident report (14 Oct 2026) · boxes = components Google names; placement and arrows are ours
other regions and Google Cloud services unaffected (per Google) · successful writes also much slower · early updates:
some reads slower / errors (not in final report) · source: status.cloud.google.com/incidents/mSFVt6PiP557bjRyYsut

Systems, Drawn · CC BY 4.0 · src/topology_oct8_bypassed.d2

4. 8 Oct, step 1: emergency capacity, blocked

Google Cloud Storage · us-central1 · 8 Oct 2026 · step 1

the first mitigation step Google lists · time NOT published

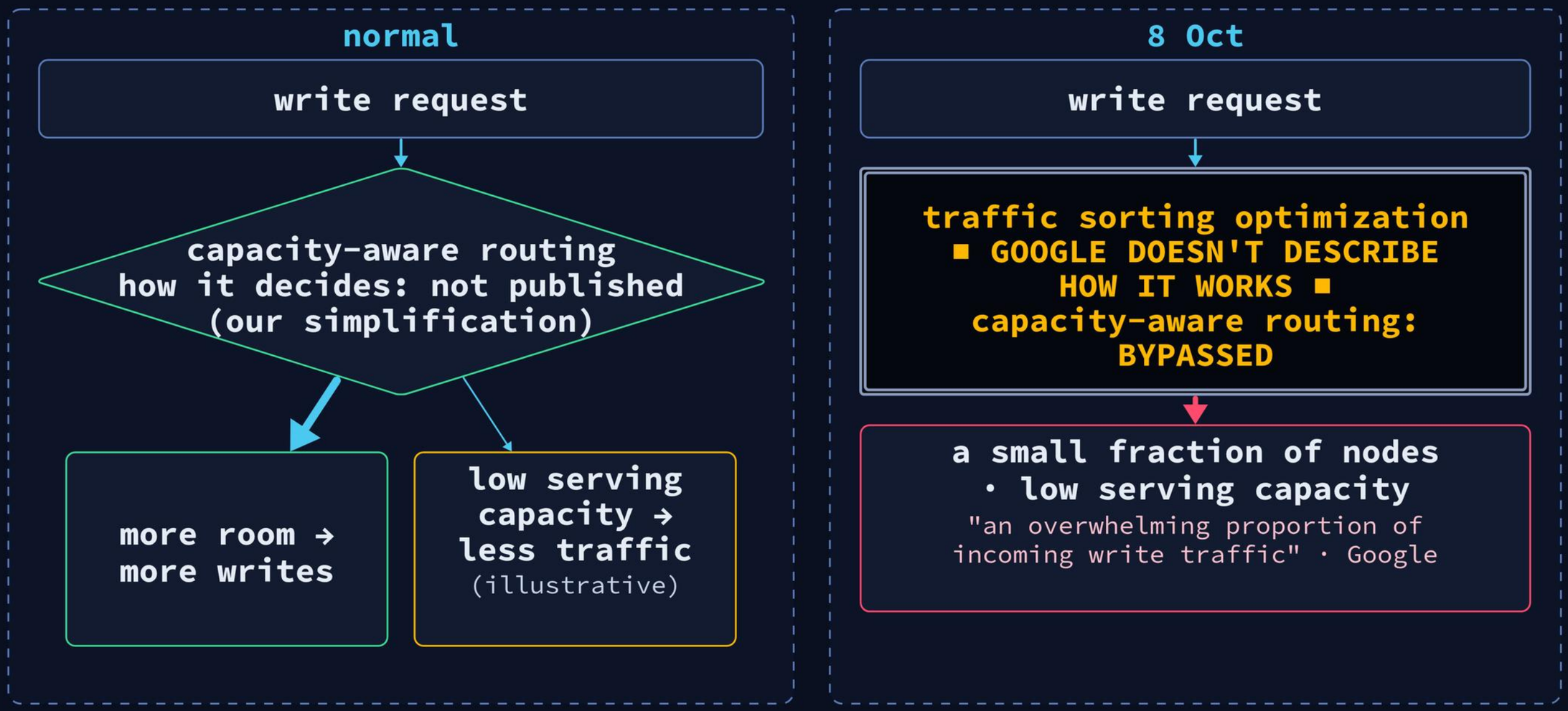


Google doesn't say what those limits were · the arrow ends at the region border: Google names the region, not nodes
source: incident report, 14 Oct 2026 · status.cloud.google.com/incidents/mSFVt6PiP557bjRyYsut

Systems, Drawn · CC BY 4.0 · src/capacity_blocked.d2

5. Same write, two routing paths

Same write, two routing paths · left: normal · right: 8 Oct 2026



Google doesn't publish how capacity-aware routing decides; the diamond is our simplification · arrow weights are illustrative
quotes: Google incident report, 14 Oct 2026 · status.cloud.google.com/incidents/mSFVt6PiP557bjRyYsut

Systems, Drawn · CC BY 4.0 · src/routing_decision.d2

6. 8 Oct morning: start, detection, first status-page update

Thu 8 Oct 2026 · morning · times PT and ART

Google Cloud Storage, us-central1 · elevated write latency and write error rates (per Google)

12 h 30 min
8 Oct · 04:00→16:30 PT
us-central1 region

11 h 35 min
between windows (our arithmetic)
no impact reported by Google

5 h 41 min
9 Oct 04:05→09:46 · zone us-central1-b

04:00 PT
08:00 ART
IMPACT STARTS
both incident reports

04:50 PT
08:50 ART
DETECTED
internal telemetry (final report)

05:19 PT
09:19 ART
⚠ LIVE UPDATES SAID
start as first stated · both reports: 04:00

10:57 PT
14:57 ART
FIRST IMPROVEMENT
internal monitoring (11:55 update)

11:24 PT
15:24 ART
FIRST STATUS-PAGE UPDATE
"Workaround: None at this time"

04:50 → 11:24 PT = 6 h 34 min

detection → first status-page update (our arithmetic) · impact start → first update: 7 h 24 min

cards in order, spacing not to scale · amber dashed = the record disagrees or has a gap · next: 8 Oct afternoon →
source: status.cloud.google.com/incidents/mSFVt6PiP557bjRyYsut (live updates, preliminary and final reports)

Systems, Drawn · CC BY 4.0 · src/timeline_oct8_morning.d2

7. 8 Oct afternoon: last live update, mitigated, the four steps

Thu 8 Oct 2026 · afternoon · Google Cloud Storage, us-central1

12 h 30 min
8 Oct · 04:00→16:30 PT
us-central1 region

11 h 35 min
between windows (our arithmetic)
no impact reported by Google

5 h 41 min
9 Oct 04:05→
09:46 · zone
us-central1-b

13:48 PT · 17:48 ART
⚠ LAST LIVE UPDATE
next entry 19 h 56 min later
Google: "an update by … 16:00 PDT"
next entry: 9 Oct preliminary report

16:30 PT · 20:30 ART
FIRST WINDOW ENDS
MITIGATED
Google: "provided temporary relief"
"Incident End"

mitigation · in the order Google lists it · times NOT published

~~1 · ADD CAPACITY~~
BLOCKED
for the region ·
"systemic regional
capacity limits"

2 · DRAIN
from the most
overloaded storage
backend nodes

3 · DISABLE
config change to
disable the traffic
sorting optimization

4 · ROLL BACK
separate issue
recent update to the
metadata database
service

only step 1 is explicitly first ("initially") · 4 is a separate issue
amber dashed = the record disagrees or has a gap · as the page shows today · "permanently resolved" only after the full revert
next: 9 Oct → · source: status.cloud.google.com/incidents/mSFVt6PiP557bjRyYsut

Systems, Drawn · CC BY 4.0 · src/timeline_oct8_afternoon.d2

8. 9 Oct: the recurrence, the reports, two ways to count the hours

Fri 9 Oct 2026 · it came back · zone us-central1-b

12 h 30 min
8 Oct · 04:00→16:30 PT
us-central1 region

11 h 35 min
between windows (our arithmetic)
no impact reported by Google

5 h 41 min
9 Oct 04:05→
09:46 · zone
us-central1-b

⚠ no status update for 19 h 56 min
none mentioned the recurrence · incident still shown as 'Service disruption' (Google's label)
8 Oct 13:48 PT (17:48 ART) → 9 Oct 09:44 PT (13:44 ART)

04:05 PT
08:05 ART
BACK · us-central1-b only
(final report)

09:44 PT
13:44 ART
PRELIMINARY REPORT
marked Available · covers 8 Oct only

09:46 PT
13:46 ART
RECURRENCE ENDS
(final report)

14 Oct 00:15 PT
04:15 ART
FINAL REPORT
first mention of the recurrence on Google's public incident page

18 h 11 min in two windows (12 h 30 min + 5 h 41 min), per Google
1 day, 6 hours = wall clock (product history page)

cards in order, spacing not to scale · amber dashed = the record disagrees or has a gap · as the page shows today
wall clock 8 Oct 04:00 → 9 Oct 09:46 PT = 29 h 46 min (our arithmetic) · cause: next scene
sources: status.cloud.google.com/incidents/mSFVt6PiP557bjRyYsut · GCS product history page

Systems, Drawn · CC BY 4.0 · src/timeline_oct9.d2

9. The incomplete rollback: three names, three boxes

The incomplete rollback · three names in Google's report, drawn as three boxes

8 Oct · us-central1 region

disable traffic sorting optimization

"A configuration change was deployed to disable the traffic sorting optimization"
· Google incident report

date NOT published

fully reverted load balancing config change

"permanently resolved" when engineering "fully reverted" the change
· Google incident report

recurrence: 9 Oct · 04:05-09:46 PT us-central1-b

'upload planner' feature · cause of the recurrence, per Google

"an incomplete rollback of an upload planner feature"
· Google incident report

⚠ Google doesn't say how the three relate

not published: what the upload planner is · why the rollback was incomplete when the full revert happened

three names in Google's report · side by side · no arrows · amber dashed = the record disagrees or has a gap
source: Google incident report, 14 Oct 2026 · status.cloud.google.com/incidents/mSFVt6PiP557bjRyYsut

Systems, Drawn · CC BY 4.0 · src/incomplete_rollback.d2

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